WEEK 2 (Mon. / Thu.)

INORGANIC EXPERIMENT 1 — SESSION I

Synthesis and Characterization of $RuCl_2(PPh_3)_3$

Venue	General & Synthetic Chemistry Lab, Block S5 Level-1	**Instrument / technique**	IR, ESI-MS
Time	10:00 a.m. – 4:00 p.m. (Mon. or Thu.)	**Source**	Manual v0724 + lecture notes

$$2\,RuCl_3 \cdot x\,H_2O + 7\,PPh_3 \longrightarrow 2\,RuCl_2(PPh_3)_3 + OPPh_3 + 2\,HCl + (2\,x\text{-}1)\,H_2O$$

SAFETY — THIS SESSION

$RuCl_3 \cdot x\,H_2O$ is corrosive and severely irritating to skin and eyes. PPh_3 is an irritant, harmful if swallowed and toxic on repeated exposure. Ethanol is an irritant and highly flammable — keep it from naked flames and the hotplate. KBr is harmful if swallowed; hot silicone oil scalds. Fume cupboard, lab coat, goggles, gloves, covered shoes. **Ruthenium waste → heavy-metal waste container**; chlorinated and non-chlorinated solvents separately.

Procedure

1. Weigh **0.50 g PPh_3** into a 100 mL RBF with stirrer bar. Add **20 mL** dried absolute EtOH. Warm to **∼40 °C for 10–15 min** until dissolved. *Water bath, not oil bath, for this step.* ("Absolute" already means water-free; "dried" additionally specifies storage over molecular sieves. The manual asks for both because the complex is moisture-sensitive.)
2. Weigh **0.08 g $RuCl_3 \cdot x\,H_2O$** into a 50 mL RBF with stirrer bar. Add **20 mL** dried absolute EtOH. Dissolve with constant stirring.
3. Transfer the ruthenium solution into the PPh_3 flask with a glass dropper. Rinse all residue in with a further **5 mL** dried absolute EtOH. *Why separately? — proforma question, see below.*
4. Fit the water-cooled reflux condenser with the $CaCl_2$ guard tube.
5. **Reflux 90 min.** The manual sets a ceiling of **130 °C**; in practice **100–110 °C** gives a steady reflux. Silicone oil PMX-200, 100 cSt, rated to 300 °C. Switch from the water bath to the oil bath at this step.
6. Cool to room temperature.
7. Collect the solid on a **Hirsch funnel** by suction filtration. Wash with cold dried absolute EtOH (**2 × 3 mL**). Dry under vacuum **20–30 min**.
8. Weigh the dried product and record it. *The % yield calculation goes in the proforma, not here.*
9. Transfer **∼10 mg** into a labelled sample tube for **ESI-MS**.
10. Record the solid IR spectrum, **500–4000 cm^{-1}**, by **ATR-IR or KBr disc**, and record which method was used. *Dry the sample fully first: this is not only for the spectrum but for the stability of the material you must still have in Week 4.*

BENCH CAUTIONS

Hold the assembly firmly enough that the condenser cannot shift and let water run into the hot oil, which spits violently — but do *not* tighten a clamp onto glass that will be heated, since glass expanding against a rigid clamp cracks. Clamp at the necks, snug rather than forced. Set the bath once and leave it: changing the setpoint alters the reaction rate, and 90 min is calibrated against that rate. The product mass is only ∼0.2 g, so minimise transfers.

Do *not* dissolve the complex to run the IR — it degrades in solution. Dry it fully and run it as a solid; with KBr, close the compartment after loading. Rinse ruthenium-contaminated glassware with **DCM**, not acetone. The complex is moisture-sensitive and water can bind as an aqua ligand, so keep air exposure short.

Proforma question — observe it today. Why are the two solids dissolved in *separate* flasks? PPh_3 needs 10–15 min at ∼40 °C, whereas $RuCl_3 \cdot x\,H_2O$ dissolves at room temperature on stirring. Dissolving them apart means both are fully in solution when they meet, so the ruthenium meets a large excess of *dissolved* phosphine rather than reacting at a solid surface, and the ruthenium salt is never held warm in ethanol on its own — alcohols can reduce Ru(III). Record the dissolution times and colours so the answer rests on your own data.

Before you leave: ☐ Labelled vial (∼10 mg) at the ESI-MS submission point ☐ IR spectrum obtained, method recorded ☐ Datasheet signed by GTA

2주차 (월 / 목)

무기 실험 1 — 세션 I

$RuCl_2(PPh_3)_3$의 합성과 특성분석

장소	General & Synthetic Chemistry Lab, Block S5 Level-1	**장비**	IR, ESI-MS
시간	오전 10시 – 오후 4시 (월 또는 목)	**출처**	매뉴얼 v0724 + 강의노트

$$2\,RuCl_3\cdot x\,H_2O + 7\,PPh_3 \longrightarrow 2\,RuCl_2(PPh_3)_3 + OPPh_3 + 2\,HCl + (2\,x\text{-}1)\,H_2O$$

안전 — 이번 세션

$RuCl_3\cdot x\,H_2O$는 부식성이며 피부와 눈에 심한 자극을 준다. PPh_3는 자극성이고 삼키면 유해하며 반복 노출 시 독성이 있다. 에탄올은 자극성이고 인화성이 높으므로 화기와 핫플레이트에서 멀리 둔다. KBr은 삼키면 유해하고 뜨거운 실리콘 오일은 화상을 일으킨다. 흄후드에서 실험복, 고글, 장갑, 발등을 덮는 신발을 착용한다. **루테늄 폐기물은 중금속 폐기물 용기에** 버리고, 할로겐 용매와 비할로겐 용매는 각각 분리한다.

실험 절차

1. PPh_3 **0.50 g**을 칭량하여 교반자를 넣은 100 mL 둥근바닥플라스크에 옮긴다. 건조 무수 에탄올 **20 mL**를 가하고 **약 40 °C에서 10–15분간** 가온하여 완전히 녹인다. 이 단계는 오일배스가 아니라 water bath를 쓴다. (absolute 자체가 이미 물이 없다는 뜻이고, dried는 여기에 더해 molecular sieve로 건조 보관했음을 뜻한다. 착물이 수분에 민감하므로 매뉴얼이 두 조건을 함께 요구한다.)
2. $RuCl_3\cdot x\,H_2O$ **0.08 g**을 칭량하여 교반자를 넣은 50 mL 둥근바닥플라스크에 옮긴다. 건조 무수 에탄올 **20 mL**를 가하고 계속 교반하여 녹인다.
3. 유리 스포이드로 루테늄 용액을 PPh_3 플라스크로 옮긴다. 건조 무수 에탄올 **5 mL**로 잔류물을 헹궈 함께 옮긴다. 두 고체를 왜 따로 녹였는지는 proforma 문항이다. 아래 주의사항 항목을 볼 것.
4. $CaCl_2$ 건조관을 연결한 수냉식 환류냉각기를 플라스크에 장착한다.
5. **90분간 환류한다.** 매뉴얼상 상한은 **130 °C**이나 실제로는 **100–110 °C**에서 환류가 안정적으로 유지된다. 실리콘 오일 PMX-200, 100 cSt, 최대 300 °C. 이 단계에서 water bath를 오일배스로 바꾼다.
6. 실온까지 냉각한다.
7. **Hirsch 깔때기**로 감압 여과하여 고체를 모은다. 차가운 건조 무수 에탄올로 세척하고(**3 mL씩 2회**) 진공에서 **20–30분간** 건조한다.
8. 건조된 생성물의 질량을 재어 기록한다. 수율 계산은 여기가 아니라 proforma에 쓴다.
9. **약 10 mg**을 라벨링한 샘플 튜브에 옮겨 **ESI-MS**용으로 제출한다.
10. 고체 IR 스펙트럼을 **500–4000 cm^{-1}** 범위에서 **ATR-IR 또는 KBr disc**로 측정하고, 어느 방법을 썼는지 기록한다. 측정 전에 완전히 건조한다. 스펙트럼 품질만의 문제가 아니라, 4주차까지 남겨 두어야 할 시료의 안정성이 걸린 문제다.

실험대 주의사항

냉각기가 밀려 냉각수가 뜨거운 기름으로 흘러들지 않도록 단단히 고정하되, 가열되는 유리에 클램프를 직접 조여서는 안 된다. 유리가 팽창하면서 고정된 클램프에 눌려 깨진다. 목 부분을 힘으로 조이지 않고 알맞게 물리는 것이 요령이다. 배스 온도는 한 번 맞춘 뒤 그대로 둔다. 도중에 바꾸면 반응 속도가 달라지고, 90분은 그 속도를 전제로 정해진 값이다. 생성물은 0.2 g 남짓이므로 이동 횟수를 최소화한다.

IR 측정을 위해 착물을 녹이지 않는다. 용액에서 분해되므로 완전히 건조한 고체로 측정하며, KBr을 쓰면 시료를 넣은 뒤 시료실을 닫는다. 루테늄이 묻은 유리기구는 아세톤이 아니라 **DCM**으로 헹군다. 착물은 수분에 민감하고 물이 aqua 리간드로 배위할 수 있으므로 공기 노출을 짧게 한다.

proforma 문항 --- 오늘 관찰할 것. 두 고체를 왜 따로 녹이는가? PPh_3는 약 40 °C에서 10–15분 가온해야 녹지만 $RuCl_3\cdot x\,H_2O$는 실온 교반만으로 녹는다. 따로 녹이면 둘이 만나는 시점에 모두 완전히 용해된 상태이므로 루테늄이 고체 표면이 아니라 용해된 과량의 포스핀과 만나며, 루테늄염만 에탄올에서 가온한 채 방치하지 않게 된다. 알코올은 Ru(III)를 환원시킬 수 있다. 용해 시간과 색 변화를 직접 기록해 관찰에 근거해 답한다.

퇴실 전 확인: ☐ 라벨링한 바이알(약 10 mg)을 ESI-MS 제출대에 제출 ☐ IR 스펙트럼 획득 및 측정법 기록 ☐ 데이터시트에 GTA 서명

DATASHEET — INORGANIC EXPERIMENT 1 — WEEK 2

Session I: Synthesis and Characterization of $RuCl_2(PPh_3)_3$

Full name (as on student ID)	Matriculation no.	Date
Lab session ☐ Mon ☐ Thu	Fumehood no.	S/N no. assigned by lab officer

1. Reagents charged

Reagent	Nominal	Actual	Remarks
PPh_3	0.50 g		
$RuCl_3 \cdot x\,H_2O$	0.08 g		
Dried absolute EtOH (to PPh_3)	20 mL		
Dried absolute EtOH (to $RuCl_3$)	20 mL		
Dried absolute EtOH (rinse)	5 mL		
Cold dried EtOH (wash, 2 ×)	3 mL		

2. Reaction conditions

	Nominal	Actual	Remarks
Warming step 1: temp. / duration	∼40 °C, 10–15 min		
Reflux start / end time	—		
Reflux duration	90 min		
Oil bath temp. / hotplate setting	< 130 °C		
Vacuum drying duration	20–30 min		

3. Product

	Value	Remarks
Mass of empty vial / paper		
Mass of vial + dried product		
Mass of $RuCl_2(PPh_3)_3$		
Colour of product		
Mass submitted for ESI-MS		
IR method ☐ ATR ☐ KBr		

4. Observations

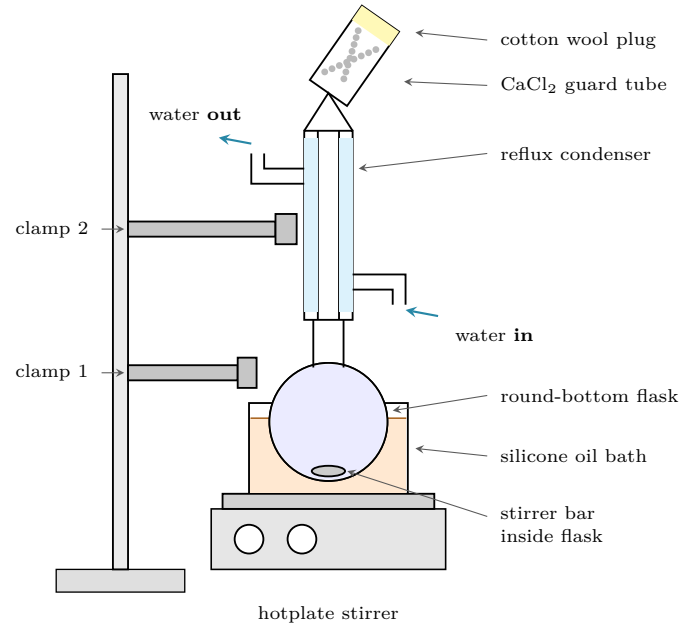
Record colour changes with the time at which they occurred: the two starting solutions, on mixing, during reflux, on cooling, of the filtrate and of the washings.

To be completed at the end of the session		
Student signature	GTA name	GTA signature

Raw data and observations must be signed by the GTA before leaving the laboratory. This datasheet is submitted as an appendix to the report.

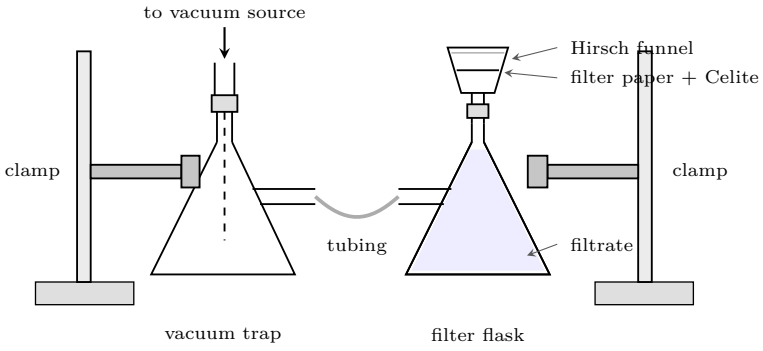
Appendix A — Apparatus Setup

A1. Moisture-free reflux assembly



Three things this drawing is meant to fix. Cooling water enters at the *bottom* and leaves at the *top*, so the jacket stays full; reversing it leaves an air pocket and the condenser stops working. **Clamp 2** holds the condenser and **clamp 1** holds the flask neck, so that neither rests its weight on the oil bath. Both grip the *necks*, and they are snug rather than forced: glass that is being heated expands, and a rigid clamp bearing on it will crack it. The grip must still be secure, because a condenser that slips lets cooling water run into the hot oil. The guard tube is packed with $CaCl_2$ and plugged with cotton wool; it excludes moisture while still venting, so it must never be sealed.

A2. Suction filtration with a vacuum trap

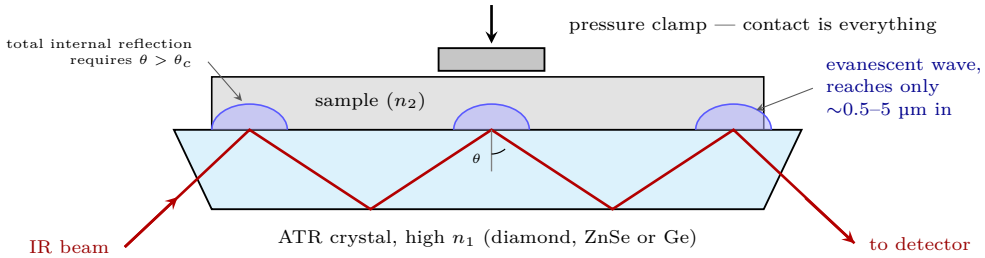


Both flasks must be clamped. A filter flask under vacuum tips easily once tubing is attached, and thick-walled glass under reduced pressure fails violently. The trap sits between the flask and the vacuum line so that liquid drawn over cannot reach the pump; it must be *empty* before use, and its inner tube must extend below the side arm or it will not trap anything. For Session I the funnel carries filter paper only; for Sessions II and III a thin layer of Celite is packed on top of the paper *with the suction already running*, so the pad does not lift.

요약. 냉각수는 아래로 들어가 위로 나가야 재킷이 물로 가득 찬다. 반대로 연결하면 공기층이 생겨 냉각이 되지 않는다. 클램프는 플라스크 목과 냉각기를 각각 잡아 무게가 오일배스에 실리지 않게 하되, 가열되는 유리를 힘으로 조이지 않는다. 유리가 팽창하면서 깨진다. 그렇다고 느슨하면 냉각기가 밀려 물이 기름으로 흘러들므로 알맞게 물리는 것이 요령이다. 건조관은 수분을 막되 압력은 빠져나가야 하므로 절대 밀봉하지 않는다. 감압 여과에서는 여과 플라스크와 트랩을 모두 클램프로 고정하며, 트랩의 안쪽 관은 반드시 곁가지보다 아래까지 내려와 있어야 한다.

Appendix C — ATR-IR: How It Works and What It Changes

C1. Total internal reflection and the evanescent wave



The beam is not passed *through* the sample. It is trapped inside a crystal of high refractive index by total internal reflection, which requires the angle of incidence to exceed the critical angle, $\sin\theta_c = n_2/n_1$. At each bounce a standing field — the **evanescent wave** — leaks a short distance into whatever is pressed against the crystal and is attenuated wherever the sample absorbs. The penetration depth is

$$d_p = \frac{\lambda}{2\pi n_1 \sqrt{\sin^2\theta - (n_2/n_1)^2}},$$

so the beam samples only the first few micrometres, and d_p **grows with wavelength**. Low-wavenumber bands are therefore probed more deeply and appear relatively stronger than in a transmission spectrum — an ATR spectrum and a library transmission spectrum of the same compound are not superimposable without correction.

C2. ATR against KBr disc — choose, then record which you chose

	ATR	**KBr disc**
Preparation	None. Place the dry solid on the crystal and clamp.	Grind with dry KBr, press a pellet.
Sample state	Native solid, undiluted.	Diluted in a salt matrix.
Atmosphere	Open bench — CO_2 near 2360 cm^{-1} and water vapour appear.	Enclosed compartment, can be purged.
Intensities	Low-wavenumber bands enhanced ($d_p \propto \lambda$).	True transmission; comparable with libraries.
Main risk	Poor contact gives a weak spectrum; crystal must be cleaned between samples.	KBr is hygroscopic — a broad O–H band near 3400 cm^{-1} is contamination, not sample.
Suits	Thick, strongly absorbing or moisture-sensitive solids.	Reference-quality spectra for comparison.

Working practice. Record a fresh background immediately before the sample, which suppresses most of the atmospheric CO_2 and water. Tighten the clamp until the signal stops improving — with a powder, most weak ATR spectra are contact failures rather than sample problems. Clean the crystal with a solvent that actually dissolves the residue: for the ruthenium complex that means **DCM**, not acetone. Run the sample *dry*; do not dissolve it, since the complex degrades in solution.

C3. What the 500–4000 cm^{-1} window can and cannot tell you

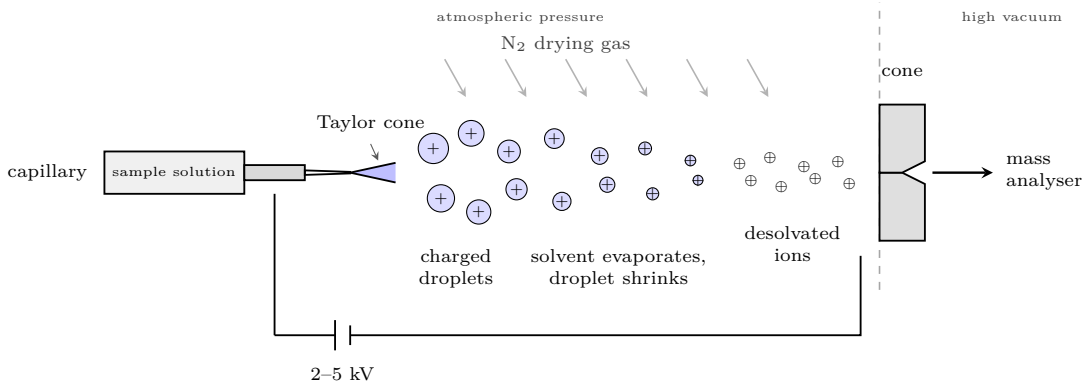
Region / cm^{-1}	**Origin**	**What it tells you**
~3050	Aromatic C–H stretch of the phenyl rings	Phosphine present
~1480, ~1435	Phenyl ring C=C; the lower band is P-phenyl in character	Phosphine present; small shifts on coordination
~1190	P=O stretch of $OPPh_3$	**Should be absent.** Its appearance means the phosphine oxide by-product was not washed out
~1090	X-sensitive phosphine mode	Phosphine present
~745, ~693	Out-of-plane C–H and ring deformation of a monosubstituted ring	Confirms the phenyl substitution pattern
2360	Atmospheric CO_2	**Not your sample.** An open-air ATR artefact

The limitation worth stating in the report. Every band above belongs to the *ligand*. The metal–ligand stretches — Ru–Cl near 300 cm^{-1} and Ru–P lower still — lie below the 500 cm^{-1} cut-off of this measurement, so the spectrum establishes that triphenylphosphine is present and that phosphine oxide is not, but it does not by itself prove that the phosphine is coordinated to ruthenium. That evidence comes from ESI-MS and from the isotope envelope. Treating the IR as confirmation of the *complex* rather than of the *ligand* is the most common overstatement in this report.

요약. ATR은 빛을 시료에 통과시키지 않는다. 굴절률이 높은 결정 안에서 전반사시키고, 반사할 때마다 결정 밖으로 스며 나오는 evanescent wave가 시료 표면 수 마이크로미터만 훑는다. 침투 깊이가 파장에 비례하므로 저파수 흡수가 투과 스펙트럼보다 상대적으로 강하게 나타나며, 라이브러리 스펙트럼과 그대로 겹치지 않는다. 개방된 상태로 측정하므로 2360 cm^{-1} 부근 CO_2는 시료가 아니라 대기이며, 측정 직전에 background를 새로 잡으면 상당 부분 상쇄된다. 스펙트럼이 약하게 나오면 대개 시료 문제가 아니라 결정과의 접촉 불량이므로 클램프를 더 조인다. 마지막으로, 이 창에서 보이는 봉우리는 전부 리간드의 것이다. Ru--Cl는 300 cm^{-1} 부근으로 측정 범위 아래에 있으므로, IR은 포스핀의 존재와 산화물의 부재를 말해 줄 뿐 배위 자체를 증명하지는 않는다.

Appendix B — How ESI-MS Works, and Why It Is Used Here

B1. From solution to gas-phase ions



The solution is pushed through a fine capillary held at 2–5 kV relative to the counter electrode. The field distorts the meniscus into a **Taylor cone** and pulls off a spray of droplets carrying excess charge. A stream of warm N_2 evaporates the solvent, so each droplet shrinks at constant charge until electrostatic repulsion overcomes surface tension and it breaks up — the **Coulombic fission** limit. Repeating this leaves bare, solvent-free ions that pass through the cone into the vacuum stage and on to the analyser.

B2. Why this matters for the interpretation

ESI IS A SOFT TECHNIQUE — THE CONSEQUENCE FOR YOUR SPECTRUM

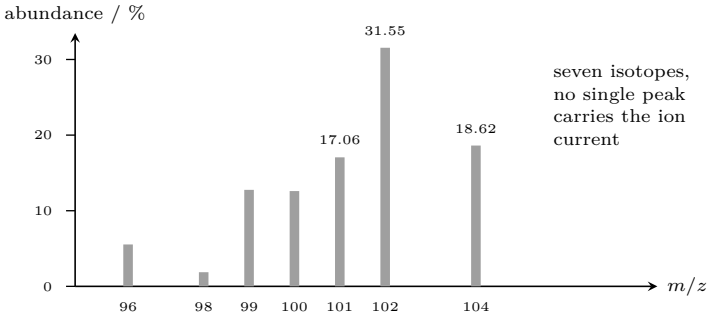
Electron ionization fires high-energy electrons at a *vapourised* sample and shatters it into a reproducible fragment pattern. ESI does none of this: there are no high-energy electrons and no energetic collisions, and the sample is never vapourised. Two things follow.

First, an involatile, thermally fragile organometallic such as $RuCl_2(PPh_3)_3$ can be analysed at all, which EI would not permit.

Second, and more useful, **any ligand loss you observe is chemistry, not instrumentation.** If the base peak corresponds to $[RuCl(PPh_3)_2]^+$ rather than the intact molecule, that reports on the lability of the apical phosphine in solution — the very same dissociation that opens a coordination site in the catalytic cycle. Under EI the same peak would carry no such meaning.

In positive-ion mode the emitter also *oxidises* the analyte. For an easily oxidised species such as ferrocene this gives the radical cation $Fc^{\bullet+}$ directly, rather than a protonated adduct.

B3. Ruthenium isotope pattern — what the cluster should look like



Ruthenium has seven stable isotopes, and chlorine adds $^{35}Cl : {}^{37}Cl = 100 : 32.5$ on top. A ruthenium chloride complex therefore produces a broad envelope of roughly a dozen peaks at unit spacing rather than one line, and **reproducing that envelope is itself evidence that the intended compound was formed**. This is the opposite of organic mass spectrometry, where C and H give weak $M{+}1$ and $M{+}2$ satellites that are easy to ignore. Predicted patterns can be generated with a free isotope distribution calculator; expect some deviation from the measured one.

One artefact to rule out. Ions formed by unimolecular processes scale with source pressure, but bimolecular association scales with its *square*. An overconcentrated sample can therefore generate dimer ions *heavier* than the molecular ion and bury it. If a peak appears at roughly twice the expected mass, dilute the sample and re-run rather than assigning it.

요약. ESI는 고에너지 전자나 충돌을 쓰지 않는 연성 이온화이므로, 스펙트럼에서 관찰되는 리간드 손실은 장비가 만든 조각이 아니라 용액에서 실제로 일어나는 화학이다. 루테늄은 안정 동위원소가 일곱 개여서 한 화합물이 십여 개의 봉우리 묶음으로 나타나며, 이 묶음의 모양이 재현되는지가 목표 화합물이 생성되었다는 증거가 된다. 시료가 너무 진하면 이분자 회합으로 분자이온보다 무거운 이온이 생겨 해석을 방해하므로, 예상보다 두 배가량 무거운 봉우리가 보이면 희석해 다시 측정한다.